# APM1205 LINEAR MODELS
# SUMMATIVE ASSESSMENT 1

## Seoul Bike Sharing Demand: Regression, ANOVA, and Model Diagnostics

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**Research question.** How are hour of day, weather conditions, season, holiday status, and functioning-day status associated with hourly rented-bike counts in Seoul?

**Dataset.** Seoul Bike Sharing Demand, UCI Machine Learning Repository. The file contains hourly observations with rental counts and weather/calendar predictors. Analysis was conducted on rows complete for the variables used.

## Data preparation and descriptive overview

| Item | Result |
|---|---|
| Raw observations | 8,760 |
| Complete analysis observations | 8,760 |
| Excluded for missing required fields | 0 |
| Rental count mean (SD) | 704.60 (645.00) |
| Rental count median (IQR) | 504 (191–1065) |
| Temperature mean (SD) | 12.88 (11.94) °C |

The analysis uses rented-bike count as the response. Hour and continuous weather measurements enter the regression as numeric predictors; season, holiday, and functioning-day status are categorical predictors. Reference categories are Winter for season and the first observed level for the other factors.

## Exploratory data analysis

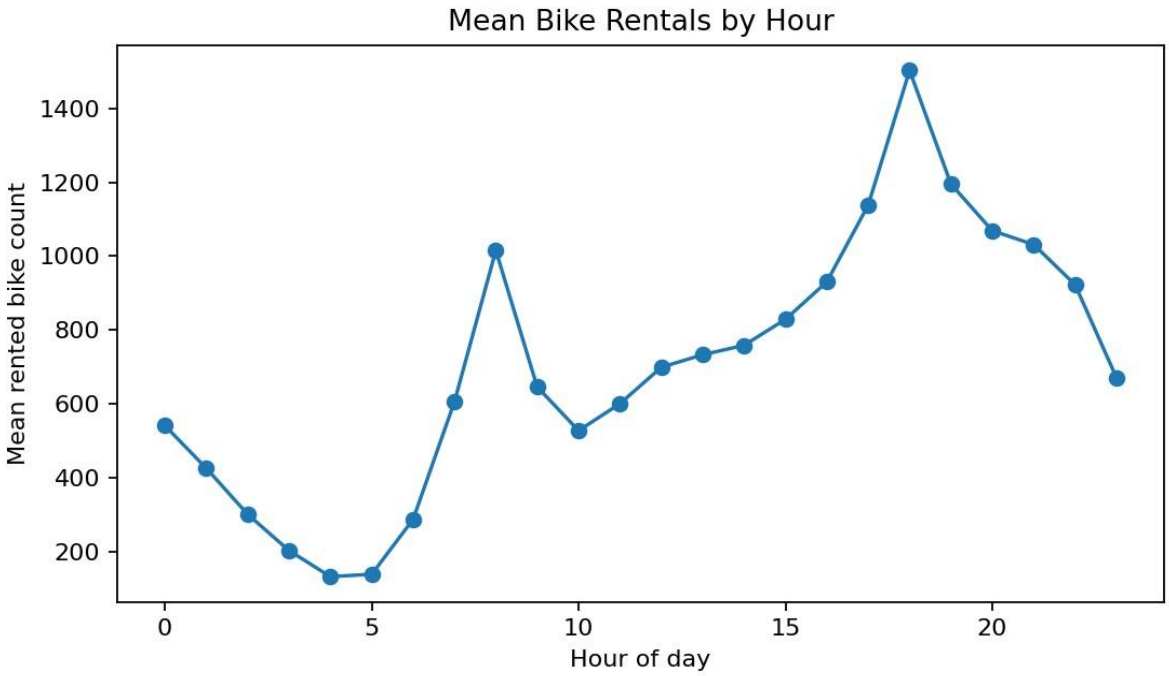



Figure 1. Mean rentals by hour of day.

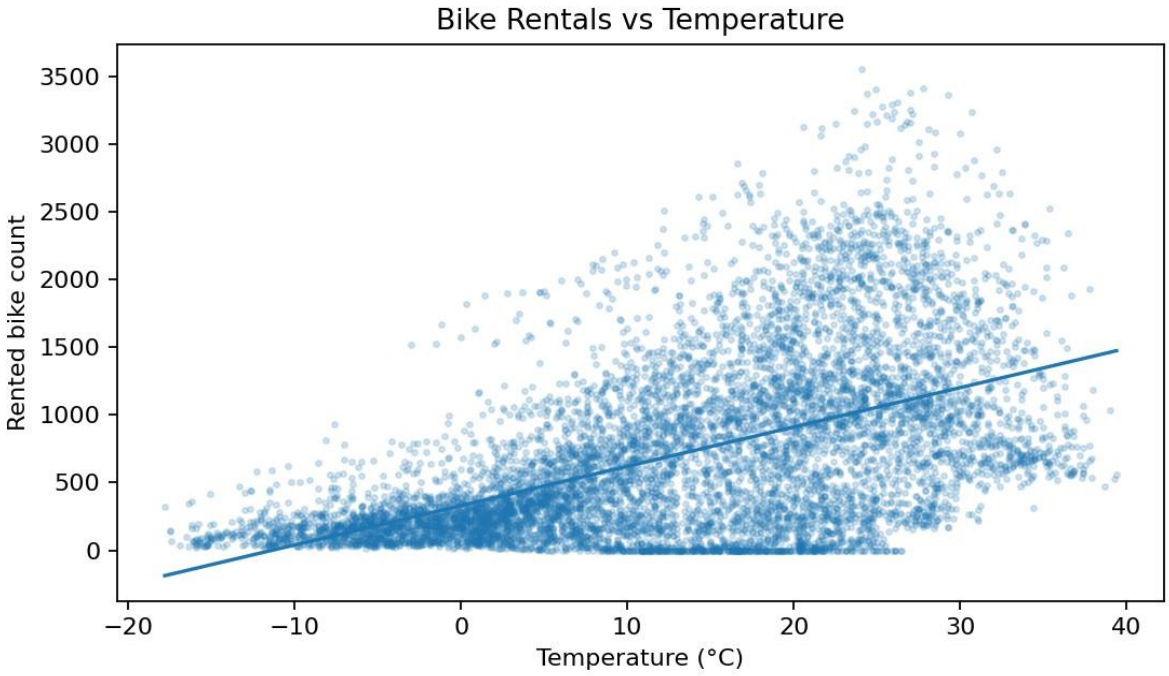



Figure 2. Rental counts versus temperature, with a fitted straight line.

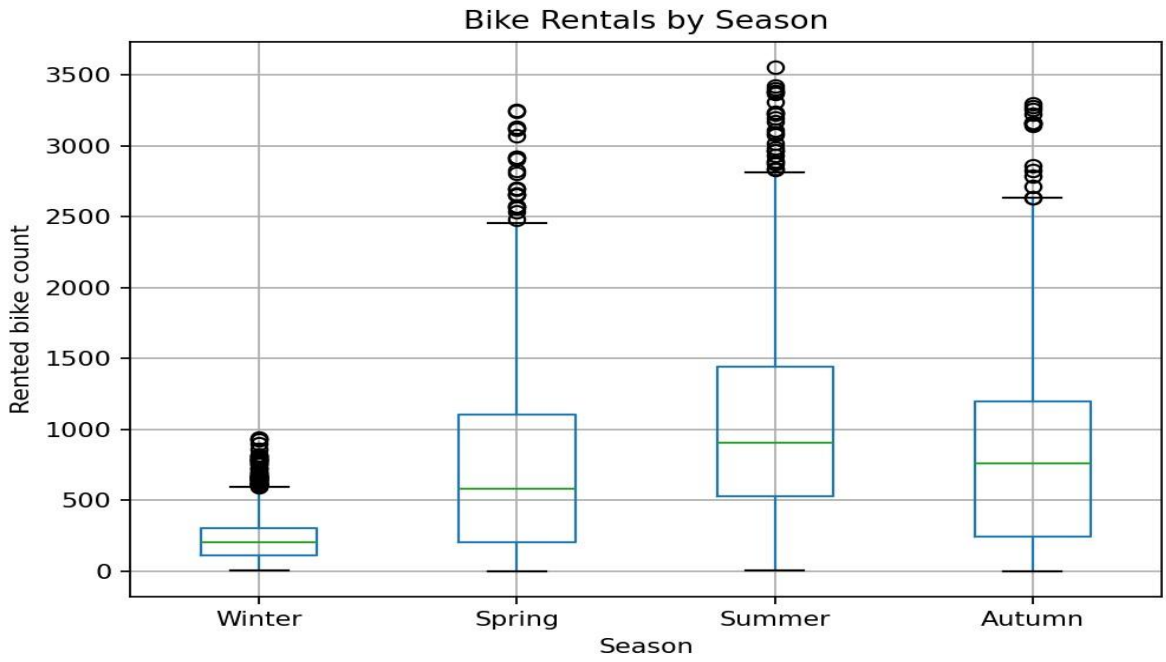



Figure 3. Distribution of rental counts by season.

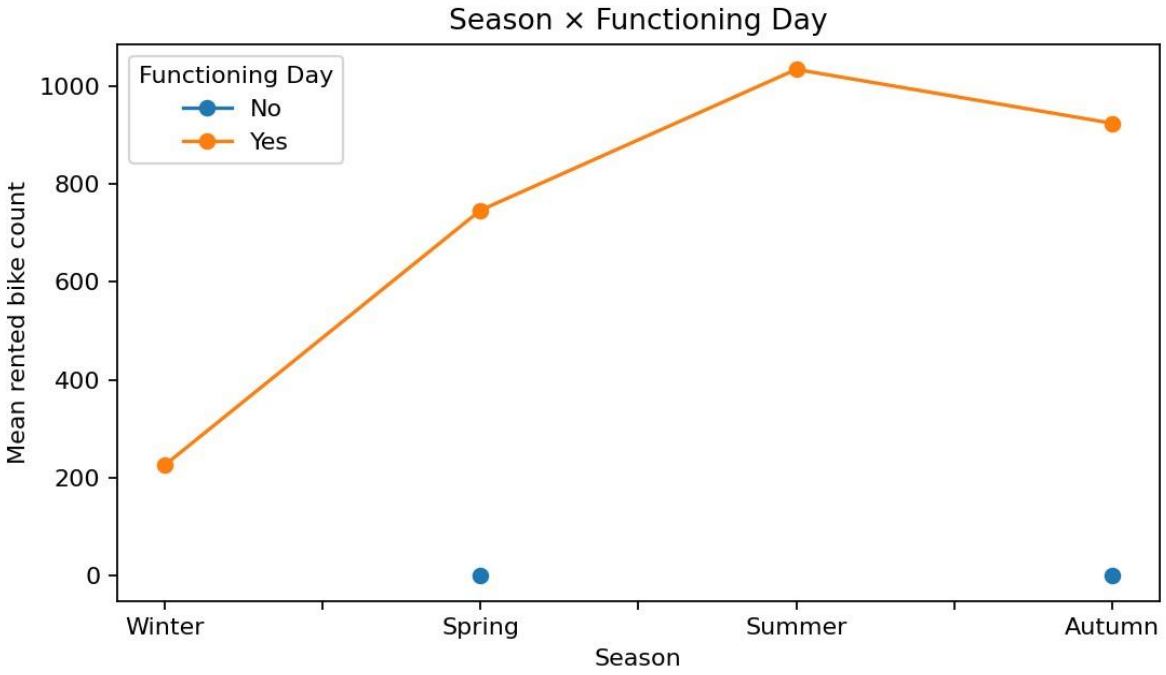



Figure 4. Mean rental count by season and functioning-day status.

The hour plot shows how demand varies across the daily cycle, while the temperature plot provides a first look at a possible linear association. Seasonal boxplots compare location and spread. The interaction plot checks whether seasonal mean differences look similar across functioning-day categories; visual patterns should be interpreted alongside model uncertainty.

## Simple linear regression

Model: rented-bike count = b0 + b1(temperature) + error. The fitted equation is:

**Predicted rentals = 329.953 + (29.081) x Temperature**

| Term | Estimate | 95% CI | p-value |
|---|---|---|---|
| Intercept | 329.953 | [313.210, 346.695] | — |
| Temperature (°C) | 29.081 | [28.128, 30.034] | 0 |

The simple model $R^2$ is 0.290 (adjusted $R^2$ 0.290). The temperature coefficient estimates the average change in rentals associated with a 1°C increase in the unadjusted linear model. Its confidence interval and p-value quantify uncertainty under the model assumptions; this association is not causal.

## Multiple regression and inference

The multiple regression adjusts for hour, temperature, humidity, wind speed, visibility, dew-point temperature, solar radiation, rainfall, snowfall, season, holiday, and functioning-day status. Coefficients are conditional associations holding the other included predictors constant.

| Model summary | Value |
|---|---|
| $R^2$ | 0.5504 |
| Adjusted $R^2$ | 0.5497 |
| Overall F statistic | 764.81 |
| Overall model p-value | 0 |
| Holdout RMSE (20%) | 433.35 rentals |
| Holdout MAE (20%) | 323.20 rentals |

| Selected term | Estimate | 95% CI | p-value |
|---|---|---|---|
| Intercept | -448.102 | [-639.168, -257.035] | 4.34e-06 |
| Hour | 27.530 | [26.090, 28.970] | 4.1e-285 |
| Q("Temperature(°C)") | 16.071 | [8.892, 23.249] | 1.16e-05 |
| Q("Humidity(%)") | -10.809 | [-12.827, -8.791] | 1.26e-25 |
| Q("Wind speed (m/s)") | 19.204 | [9.216, 29.192] | 0.000165 |
| Q("Solar Radiation (MJ/m2)") | -77.310 | [-92.194, -62.426] | 3.27e-24 |
| Q("Rainfall(mm)") | -58.480 | [-66.851, -50.109] | 2.98e-42 |

Use the coefficient table CSV for all model terms. Because temperature and dew point, and other weather measures, may be correlated, individual coefficient signs and p-values should be interpreted with caution. The large sample can make small effects statistically significant, so practical magnitude also matters.

## Least-squares matrix formulation

The multiple regression can be written as y = Xb + error. Here y is the n x 1 vector of rental counts and X is the n x p design matrix containing a column of ones, numeric predictors, and dummy-coded factor levels. The least-

squares estimator minimizes ||y - Xb||^2; when X has full column rank, b_hat = inverse(X_transpose X) X_transpose y. In practice, stable QR-based methods are preferred to explicitly inverting the cross-product matrix.

| Matrix property | Value |
|---|---|
| n (analysis rows) | 8,760 |
| p (design columns) | 15 |
| Rank of X | 15 |
| Full column rank | Yes |

## Dummy variables and ANOVA

For season-only ANOVA, the null hypothesis is that all four seasonal mean rental counts are equal. In dummy-variable regression, the intercept represents the reference season mean and each season coefficient compares that season with the reference season, Winter. The two-way model includes season, functioning-day status, and their interaction.

| Season-only ANOVA | df | F | p-value |
|---|---|---|---|
| Season | 3 | 776.47 | 0 |

| Two-way ANOVA component | df | F | p-value |
|---|---|---|---|
| Season × Functioning Day | 3 | 0.00 | 1 |

A small interaction p-value would indicate evidence that the seasonal pattern differs by functioning-day status. If not small, the data do not provide strong evidence of differing seasonal effects under this model. ANOVA conclusions depend on the usual linear-model assumptions.

## Diagnostics and validation

The Breusch–Pagan test for nonconstant residual variance gives LM = 844.31, p = 3.65e-171. A small p-value suggests heteroscedasticity, so classical standard errors and confidence intervals may be unreliable. Inspect residual-versus-fitted and Q–Q plots from the R script; also examine leverage and influential points. Multicollinearity is plausible among temperature, dew point, and related weather variables, so check variance inflation factors from the R output.

For a reproducible holdout check, an 80/20 random split with seed 1205 produced RMSE = 433.35 and MAE = 323.20 rentals on the held-out subset. RMSE penalizes large errors more heavily; MAE is the average absolute error. Because observations are hourly and temporally ordered, a random split may be optimistic; a time-based split is a useful additional validation.

## Conclusion and limitations

The fitted multiple regression explains 55.0% of the variation in the analyzed rental counts in-sample. Hour and weather/calendar variables jointly provide information about rental demand, but a linear model cannot capture every nonlinear relationship or temporal pattern. The final model should be justified using fit, interpretability, diagnostic evidence, and holdout error—not $R^2$ alone.

Limitations include observational data, possible omitted variables, nonlinear weather effects, temporal autocorrelation, nonconstant residual variance, and potential multicollinearity. Results describe associations and should not be interpreted as causal effects. Check all outputs by running code/SANTOS_SA1.R in R before submission.

Dataset source: UCI Machine Learning Repository, Seoul Bike Sharing Demand: https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand